

An Explainable Data-Driven Platform for Optimal Wind Farm Siting

Francesco Bizzarri

Student ID:
35740639

Supervisors:
Tim Dwyer (Monash University)
Simone Marinai (University of Florence)

Submitted for FIT5126 as an exchange student at Monash University, May 2025

1 Introduction

Wind farm site selection is a crucial yet highly complex challenge. Optimal siting involves much more than simply identifying windy locations; it requires balancing multiple, often competing factors including, but not limited to, proximity to the electricity grid, consistency of wind resources, land use constraints, environmental concerns, community acceptance, and economic viability [1–3]. The goal is to select locations that will not only generate the most reliable energy over time but also integrate well into the broader power system and surrounding landscape. However, this decision-making process is often not systematically optimized.

This issue has become even more pressing in the context of growing climate change awareness and the global transition toward sustainable energy systems. In Australia, policy commitments such as the Victorian Renewable Energy Targets, legislated in the *Renewable Energy Act (2017)* [4], reinforce this shift. These targets include:

- 40% of Victoria’s electricity generation to be renewable by 2025
- 65% by 2030
- 95% by 2035

Meeting such goals will demand not only increased renewable energy capacity, but also smarter, more coordinated deployment strategies. This underscores the need for data-driven and holistic planning tools that can guide decision-making.

Currently, wind farm development in Australia is largely driven by a bottom-up planning ap-

proach. While site selection occurs within a defined legal framework and requires approval from energy market regulators, final decisions are often shaped by the interests of local stakeholders, like landowners, and by wind resource assessments conducted over relatively short timescales. Given that wind farms typically operate for 20 to 30 years, this approach can result in long-term inefficiencies [5]. Studies have shown that such a fragmented planning process can lead to suboptimal outcomes in terms of both total energy generation and supply stability. In contrast, a top-down planning model, where wind farm locations are determined with long-term demand patterns in mind, has demonstrated the potential for significantly greater efficiency and reliability [6].

1.1 Aim

This literature review aims to explore existing methods for wind farm site selection, both globally and within Australia. It will examine traditional approaches as well as recent techniques that incorporate artificial intelligence, with a particular focus on how explainable these methods are, both in terms of technical transparency (e.g., explainable AI) and in their ability to clearly justify decisions to non-technical audiences.

Findings from the review will inform the design of a user-friendly, explainable decision-support platform intended for a diverse range of users. These can include both technical actors such as infrastructure planners, power operators (e.g., AEMO, VicGrid) and decision makers, as well as ordinary citizens and local communities. In this context, clear and interactive representations of site selection criteria can help build trust, reduce misunderstandings, and support more inclu-

sive planning. This is particularly important because the question of where to produce renewable electricity is often controversial, and the acceptance of new energy infrastructures, especially by the communities living near proposed sites, is critical, and it can lead to tension between stakeholders with different priorities or levels of technical understanding. [2, 7–9].



Figure 1: Recent protests against wind farms in Australia. **(Center)** A protest in Wollongong against plans for an offshore wind farm, May 2025 (Photo: Anna Warr, The Advocate). **(Right)** A rally on the front lawns of Parliament House in Canberra, February 24 (Photo: Mike Bowers, The Guardian). **(Left)** A sign photographed by the author in Warnambool, March 2025

1.2 Document structure

Following the introduction, this report presents a structured literature review divided into several sections. This review critically examines traditional and AI-based methods for wind farm site selection, with a particular focus on how explainable the methods are. It includes comparisons between approaches used across different countries, as well as a section dedicated to Australian visual and interactive tools, mainly focusing on mapping systems. These tools, coming from different domains than wind energy planning, offer inspiration and practical features that will help shape the design of the proposed platform. The State of the Art Summary then identifies gaps in the existing literature, particularly the lack of explainable and accessible visual tools that allow the creation of “what-if” scenarios. The Research Project Plan outlines a practical approach to addressing these gaps, including a section on the technical tools to be used, such as programming languages and frameworks. Finally, the Conclusion section highlights the key findings and provides a concise summary of the report.

2 Literature review

2.1 Data Inputs for Wind Farm Site Suitability Modeling

Before delving into the various methods used for wind farm site selection, it is important to first understand the types of data commonly employed in these studies. The selection of suitable wind farm sites is inherently data-driven, relying on a wide range of datasets to evaluate the feasibility, efficiency, and sustainability of potential locations. These datasets, which span technical, environmental, and social factors, are fundamental in determining the most appropriate sites for wind farm development. After reviewing numerous studies it becomes evident that the types of data used are similar, regardless of geographical location or the specific approach employed. The most common siting factors identified across these studies are summarized in the table below:

Siting Factor
Wind Speed
Distance to Protected or Wildlife Areas
Slope
Distance to Urban Centers
Distance to Roads
Distance to Transmission Lines or Substations
Life Cycle Costs
Distance to Airports
Distance to Water Bodies
Elevation
Distance to Animal Habitats or Migration Zones
Wind Power Density
Distance to Agricultural Areas
Flood Risk
Landslide Risk
Distance to Historic Places
Distance to Railroads
Distance to Military Zones

Table 1: Common siting factors used in (onshore) wind farm site selection

However, even with these shared factors, a significant challenge persists: the lack of standardization in how data is represented and utilized. As highlighted in [10], a consistent and standardized approach to data representation is absent across the studies. This variability in data handling can create discrepancies in the results and undermines the reliability of findings across different studies,

especially if applied to the same region of interest. The wind speed, for example, may be categorized either as an economic factor [11, 12], or as a technical factor [13, 14]; as a constraint [15] (setting a minimum threshold) or as an evaluation criterion [16] (taken into account in order to maximize potential energy output). Furthermore, the choice of data sources often remains unclear, with many studies failing to specify where their datasets were obtained [1, 13, 17–20], which hinders transparency and replicability in site selection research.

References

- [1] S. Ali, S.-M. Lee, and C.-M. Jang, “Determination of the most optimal on-shore wind farm site location using a gis-mcdm methodology: evaluating the case of south korea,” *Energies*, vol. 10, no. 12, p. 2072, 2017.
- [2] M. Drechsler, J. Egerer, M. Lange, F. Masurowski, J. Meyerhoff, and M. Oehlmann, “Efficient and equitable spatial allocation of renewable power plants at the country scale,” *Nature Energy*, vol. 2, no. 9, pp. 1–9, 2017.
- [3] J. Huenteler, T. Tang, G. Chan, and L. D. Anadon, “Why is china’s wind power generation not living up to its potential?” *Environmental Research Letters*, vol. 13, no. 4, p. 044001, 2018.
- [4] State of Victoria, “Renewable Energy (Jobs and Investment) Act 2017 (Vic),” Retrieved from: <https://www.legislation.vic.gov.au/in-force/acts/renewable-energy-jobs-and-investment-act-2017/002>, 2017.
- [5] J. M. Weinand, E. Naber, R. McKenna, P. Lehmann, L. Kotzur, and D. Stolten, “Historic drivers of onshore wind power siting and inevitable future trade-offs,” *Environmental Research Letters*, vol. 17, no. 7, p. 074018, 2022.
- [6] A. Gunn, R. Dargaville, C. Jakob, and S. McGregor, “Spatial optimality and temporal variability in australia’s wind resource,” *Environmental Research Letters*, vol. 18, no. 11, p. 114048, 2023.
- [7] G. S. Nieminen and E. Laitinen, “Understanding local opposition to renewable energy projects in the nordic countries: A systematic literature review,” *Energy Research & Social Science*, vol. 122, p. 103995, 2025.
- [8] L. Susskind, J. Chun, A. Gant, C. Hodgkins, J. Cohen, and S. Lohmar, “Sources of opposition to renewable energy projects in the united states,” *Energy Policy*, vol. 165, p. 112922, 2022.
- [9] C.-L. Shen and H.-S. Tai, “National goal, local resistance: How institutional gaps hinder local renewable energy development in taiwan,” *Energy for Sustainable Development*, vol. 83, p. 101586, 2024.
- [10] J. J. Wimhurst, C. C. Nsude, and J. S. Greene, “Standardizing the factors used in wind farm site suitability models: A review,” *Heliyon*, vol. 9, no. 5, 2023.
- [11] H. Eroğlu, “Multi-criteria decision analysis for wind power plant location selection based on fuzzy ahp and geographic information systems,” *Environment, Development and Sustainability*, vol. 23, no. 12, pp. 18278–18310, 2021.
- [12] D. J. Pojadas and M. L. S. Abundo, “Spatio-temporal assessment and economic analysis of a grid-connected island province toward a 35% or greater domestic renewable energy portfolio: a case in bohol, philippines,” *International Journal of Energy and Environmental Engineering*, vol. 12, pp. 251–280, 2021.
- [13] B. A. Ajanaku, M. P. Strager, and A. R. Collins, “Gis-based multi-criteria decision analysis of utility-scale wind farm site suitability in west virginia,” *GeoJournal*, vol. 87, no. 5, pp. 3735–3757, 2022.
- [14] P. Díaz-Cuevas, “Gis-based methodology for evaluating the wind-energy potential of territories: A case study from andalusia (spain),” *Energies*, vol. 11, no. 10, p. 2789, 2018.
- [15] M. S. Genç, F. Karipoğlu, K. Koca, and Ş. T. Azgın, “Suitable site selection for offshore wind farms in turkey’s seas: Gis-mcdm based approach,” *Earth Science Informatics*, vol. 14, no. 3, pp. 1213–1225, 2021.
- [16] A. Ouammi, V. Ghigliotti, M. Robba, A. Mimet, and R. Sacile, “A decision support system for the optimal exploitation of

- wind energy on regional scale,” *Renewable Energy*, vol. 37, no. 1, pp. 299–309, 2012.
- [17] J. Kazak, J. Van Hoof, and S. Szewranski, “Challenges in the wind turbines location process in central europe—the use of spatial decision support systems,” *Renewable and Sustainable Energy Reviews*, vol. 76, pp. 425–433, 2017.
- [18] M. Kabak and G. Taşkınöz, “Determination of the installation sites of wind power plants with spatial analysis: A model proposal,” *Sigma Journal of Engineering and Natural Sciences*, vol. 38, no. 1, pp. 441–457, 2020.
- [19] A. Azadeh, A. Rahimi-Golkhandan, and M. Moghaddam, “Location optimization of wind power generation–transmission systems under uncertainty using hierarchical fuzzy dea: A case study,” *Renewable and Sustainable Energy Reviews*, vol. 30, pp. 877–885, 2014.
- [20] A. Azadeh, S. Ghaderi, and M. Nasrollahi, “Location optimization of wind plants in iran by an integrated hierarchical data envelopment analysis,” *Renewable Energy*, vol. 36, no. 5, pp. 1621–1631, 2011.